

TANYA SINGH

+1 412(780)9657 | tanyas2@alumni.cmu.edu | www.linkedin.com/in/tanya-singh-aecmanagement

Assistant Project Manager with 5+ years across various capital mission-critical programs working owner-side, with experience in cost, schedule, and stakeholder management on ground-up programs up to \$9B, spanning across 10+ buildings and complex MEP systems. Building agentic AI systems for change management workflows using deterministic algorithms for reliability, sub-second LLM components, ETL pipelines into PostgreSQL for data integrity, and strong guardrails with evaluation metrics for observability, deployable at program level.

EDUCATION

Carnegie Mellon University - MS in Architecture, Engineering, and Construction Management (MS-AECM)

Graduated: December 2025 | GPA: 3.74/4 |

Guru Gobind Singh Indraprastha University, Bachelor in Architecture

Graduated: July 2022 | GPA: 8.22/10

WORK EXPERIENCE

Mace Consult, IN

May 2026 – Present

Assistant Project Manager

- Driving improvements to change tracking workflows in Oracle Unifier and Procore, standardizing documentation, status stages, and reporting cadence to reduce cycle time and manual rework in engineering change review.
- Assess design changes and change orders against issued drawings, site conditions, and contract terms as owner's representative, determining validity and cost impact across 10+ buildings.
- Support project controls on a \$9B ground-up program: track budget variance and building recurring cost and change reports in Excel and Power BI.

Department of Architecture, Carnegie Mellon University, PA

Aug 2024 – Dec 2025

Research Assistant, Sustainable Materials & Life Cycle Assessment

- Developed Post-Occupancy Evaluation protocols converting building performance across IAQ, thermal and environmental conditions, energy, associated costs, converting the data into operational recommendations.

Hill International, Inc., PA

May – Aug 2025

Project Management Intern, Aviation & Transit Infrastructure

- Assessed design changes against issued drawings and specifications to determine scope, cost, and schedule impact as owner's representative, supported MEP commissioning coordination, AHJ approvals, and contract compliance reviews.
- Reconciled installed quantities using Bluebeam takeoff and markup, contractor payment applications, and RFIs against drawings, bid items, and specifications, maintaining the Item Book spreadsheets for accurate cost controls.
- Led innovation initiatives, procured and deployed time-lapse site monitoring systems and drafted an Envision certification framework for the aviation program.

Arcop Associates Private Limited

Aug 2021 – Jul 2024

Intern to Project Architect, Bharti Realty Mega Commercial, Aerocity, Delhi, India

- Spearheaded cross-disciplinary coordination, design, and management for a \$ 2B, 3.5M sq ft LEED-certified commercial office and retail development, coordinating architectural and engineering drawings in Revit and AutoCAD across 25+ stakeholders.
- Coordinated MEP and utilities interfaces during superstructure execution, resolving constructability and sequencing issues with consultants and contractors.
- Managed quantity-takeoffs, BOQs, progress reporting, and cost tracking workflows with QA/QC accuracy.
- Secured multi-agency regulatory approvals (DUAC, Environmental NOCs, Airport) ahead of schedule, administered change orders and cost management with project and program managers. Drove design efficiency gains through proactive MEP value engineering, and layout optimization (12%), and achieved 15% procurement savings through strategic vendor negotiation.

PROJECTS

Constant Engineering Change Management System for Capital Programs

- Built the data architecture to keep one change auditable across its whole life: A 22-entity relational schema spanning multiple contracts and programs, with row-level tenant isolation guardrail, migrated from SQLite to PostgreSQL for transactional reliability, with full-text search, vector indexes, and server-side pagination validated against a seeded register of 10,000+ change records.
- Standardized a fragmented change process into one system of record: One persistent identifier per change from origination through executed change order for traceability, a 21-status state machine with 34 enforced transitions across 11 roles, working day SLA clocks, tiered approval routing by cost threshold, and an append-only audit log for defensible reconstruction of every decision.
- Designed and built the AI entitlement assistant under strict governance: Clause retrieval with a deterministic pre-screen and explicit abstention, human-in-the-loop enforcement so automation drafts and routes but never decides, and deny-by-default column access keeping internal estimates out of prompts. Red-teamed and evaluated.

ReVIVE Condemned Property Triage Tool (MS AEC Management Thesis, Mayor's office expressed adoption interest)

- Built an automated ETL and Spatial-join pipeline unifying 11municipal and public datasets into a single 8900-parcel, 144-attribute register resolving fragmented records across the city, county, and federal sources into one queryable ground truth.
- Designed a configurable scoring and routing engine ranking 4500 condemned properties (2020-2025) against eight weighted criteria routing each to the responsible department by score threshold and capacity, with an interactive map and exportable reports.
- Extending with an AI agent layer that diagnoses stalled condemned properties, structured extraction over lien, deed and code violation records, with citation-grounded statutory retrieval, classifying the blocking condition behind 50%+unresolved cases.

SKILLS & CERTIFICATIONS

- OSHA 10 hr safety, Construction Manager in Training level I, and LEED GA (in process) certified
- Tools & Platforms - Oracle Primavera Unifier, Revit, Autodesk AutoCAD, Bluebeam, Procore, Excel and Google Sheets, SQL